

Math 104 Homework 3

Due Date: Friday, July 17 at 11:59pm

Instructions: Submit pdf solutions in LaTeX (preferred), or *neatly* handwritten. Unless otherwise stated, you may cite any result from class or from Rudin's book, but all other assertions you make must be proved rigorously. At the top of your submission, please write down the names of the other students you collaborated with and any outside sources you consulted.

1. Let (X, d) be a metric space. A subset $E \subset X$ is called *dense* in E if, for any open $O \subset X$, we have $O \cap E \neq \emptyset$. A metric space (X, d) is called *separable* if it has a countable dense subset.
 - (a) Show that $\mathbb{Q}$ is dense in $\mathbb{R}$, and conclude that $\mathbb{R}$ is separable.
 - (b) Show that a subset E is dense in X if and only if $\overline{E} = X$.
 - (c) By following these steps, prove that any compact metric space (X, d) is separable:
 - i. Using problem 5(c) on homework 2, show that for each $n \in \mathbb{N}$, the following is true: there is a finite set $S_n \subset X$ so that every point of X is within distance $\frac{1}{n}$ of some point of S_n .
 - ii. Let $S = \bigcup_{n=1}^{\infty} S_n$. Show that S is countable.
 - iii. Show that every point of X is either in S or a limit point of S . Conclude that S is dense in X , and hence that X is separable.
2. Using **only** the definitions of $\lim$, $\limsup$, and $\liminf$, prove the following statements.
 - (a) $\lim_{n \rightarrow \infty} [\sqrt{n+1} - \sqrt{n}] = 0$. *Hint:* Use difference of squares.
 - (b) $\limsup_{n \rightarrow \infty} (-2)^n = \infty$.
 - (c) $\limsup_{n \rightarrow \infty} [-\sqrt{n}] = -\infty$.
 - (d) $\liminf_{n \rightarrow \infty} \left[\frac{1}{n} + (1 + (-1)^n)n \right] = 0$.

3. Let $\{s_n\}$ be an arbitrary sequence in $\mathbb{R}$.

(a) Let $s \in \mathbb{R}$. Show that $\limsup_{n \rightarrow \infty} s_n = s$ if and only if the following two conditions hold:

- For all $\epsilon > 0$, there is $N \in \mathbb{N}$ so that $s_n < s + \epsilon$ for all $n \geq N$.
- For all $\epsilon > 0$ and for all $N \in \mathbb{N}$, there exists $n \geq N$ so that $s_n > s - \epsilon$.

(b) Show that some subsequence of $\{s_n\}$ converges (in the sense of extended real numbers, if necessary) to $\limsup_{n \rightarrow \infty} s_n$ and some subsequence of $\{s_n\}$ converges to $\liminf_{n \rightarrow \infty} s_n$.

(c) Define

$$E = \{x \in [-\infty, \infty] : \text{some subsequence } \{s_{n_k}\} \text{ of } \{s_n\} \text{ converges to } x\}.$$

Show that $\limsup_{n \rightarrow \infty} s_n = \sup E$ and $\liminf_{n \rightarrow \infty} s_n = \inf E$.

Hint: what happens to $\limsup$ and $\liminf$ on subsequences?

4. Let $a > 1$. In this problem, we will define what it means to write a^r for a positive *irrational* number r . We first define exponentiation for positive *integer* exponents in the usual way, via repeated multiplication. We know this satisfies some properties, for example

$$a^{r+s} = a^r a^s \quad (*)$$

$$(ab)^r = a^r b^r \quad (\Delta)$$

for $a, b > 0$ and $r, s > 0$ integers. Now, we proved in class that for any positive integer q , there is a *unique* positive number whose q th power is a , which we denote $\sqrt[q]{a}$. For a positive rational number $r = \frac{p}{q}$, we define $a^r = \sqrt[q]{a^p}$.

(a) Show that this definition makes sense, i.e. is independent of the representation of r as a fraction. That is, show that if rational numbers $\frac{p}{q} = \frac{r}{s}$ then $\sqrt[q]{a^p} = \sqrt[s]{a^r}$.

Hint: use lowest terms and the uniqueness of q th roots.

(b) Show that properties $(*)$ and (Δ) above still hold for positive rational numbers.

Hint: find a common denominator and use uniqueness again.

(c) For positive rational $r < s < t$, show that $a^r < a^s$ and that

$$0 < a^s - a^r = a^r (a^{s-r} - 1).$$

Conclude that for any positive rational r, s and any rational t larger than both r and s , we have

$$|a^r - a^s| < a^t (a^{|r-s|} - 1).$$

- (d) Now let r be a positive *irrational* number. We know there is a sequence of rational numbers q_k that converge to r (for example, the decimal expansion of r). We define

$$a^r = \lim_{k \rightarrow \infty} a^{q_k}.$$

Show that the sequence $\{a^{q_k}\}$ is Cauchy, and conclude that this limit exists.

Hint: Use part (c) and Rudin Theorem 3.20(d).

- (e) Show that the choice of sequence $\{q_k\}$ is irrelevant; that is, show that if $\{p_k\}$ and $\{q_k\}$ are sequences of rational numbers converging to r , then

$$\lim_{k \rightarrow \infty} a^{p_k} = \lim_{k \rightarrow \infty} a^{q_k}.$$

- (f) Using limit laws, prove that properties (*) and (Δ) continue to hold for positive irrational exponents.

Remark: For $r < 0$ we then define $a^r = \frac{1}{a^{-r}}$, and for $0 < a < 1$ we define $a^r = \left(\frac{1}{a}\right)^{-r}$. It is then a matter of checking cases to verify that all the properties of exponentiation we want are indeed true.